MCF-LP BY-NC-ND-EP · MORPHEAN COSMOLOGY FRAMEWORK

MCF Hydrogen Mesh Blueprint

Modular alkaline electrolyzer system — open-source, free to the world. Benefits stated honestly. Boundaries stated clearly. No hype.

Christos Sotirelis — "Morpheas the Shape Master" · April 2026

1. CORE DEFINITION — WHAT THIS SYSTEM IS

ONE CLEAR STATEMENT

The MCF Hydrogen Mesh is a **modular alkaline electrolysis system** that converts electrical energy into hydrogen gas for storage and later use. It is an energy conversion and storage tool. It is not an energy source. It does not create energy.

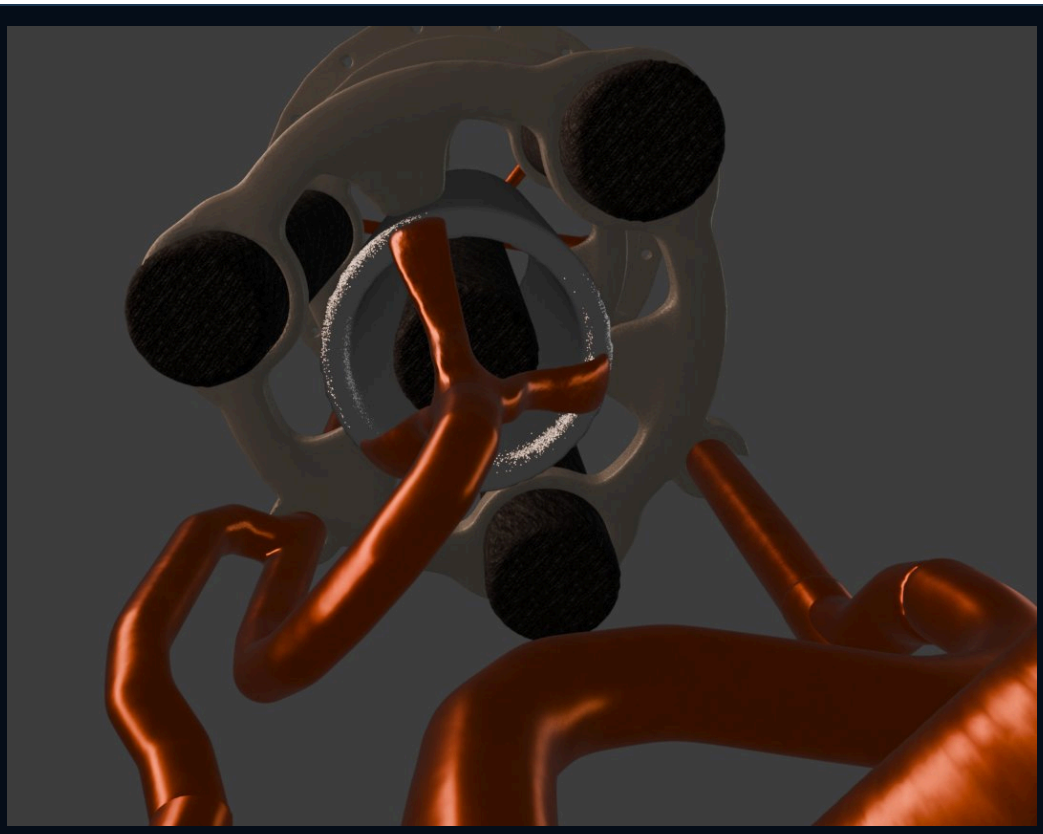


Fig. 1 — Top orthographic wireframe showing the 120° tri-symmetric electrode arrangement. Central nickel cathode surrounded by three carbon anodes.

THE PHYSICAL REACTION

$2H_2O$ + electrical energy → $2H_2$ + O_2 Electrical energy splits water at the electrodes. Hydrogen gas forms at the cathode (nickel pipe). Oxygen gas forms at the anode (carbon rods). KOH solution carries the ion current between them.

IT IS

An energy **conversion** system
An energy **storage** system (via H_2)
A **modular, scalable** architecture
A **backup and buffer** layer

IT IS NOT

An energy **creation** system
A **self-sustaining** power loop
An **energy amplifier**
A **propulsion replacement**

2. 3D DESIGN VISUALISATION — BLENDER 4.2.3 LTS

All renders below are from the original MCF Blender project file. These are the mechanical blueprints for the 300 mm glass cylinder, tri-symmetric electrode assembly, and modular column stack.

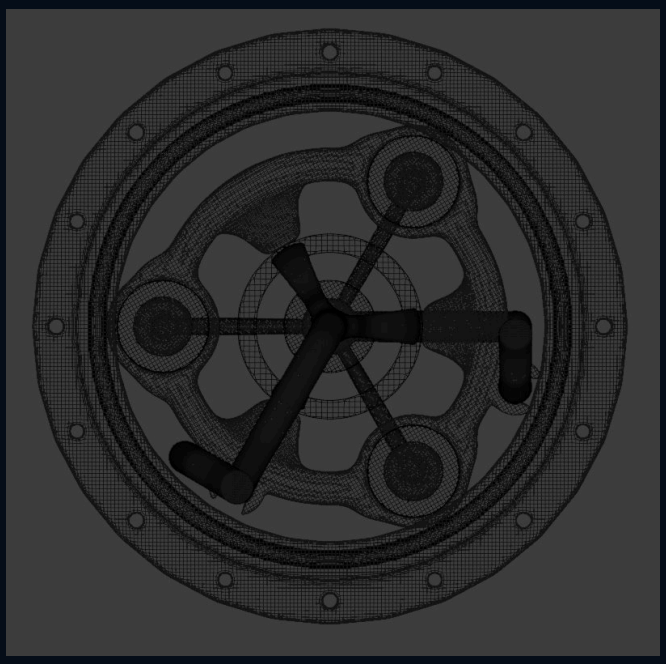


Fig. 2 — Internal electrode assembly detail

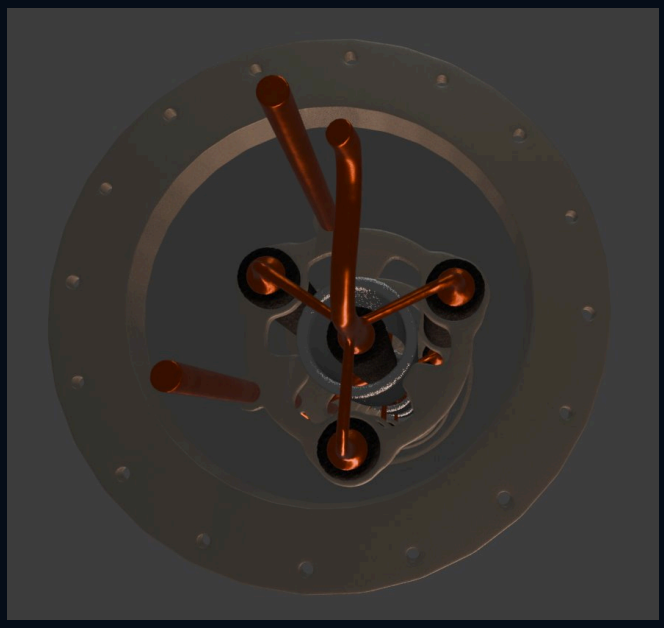


Fig. 3 — Bottom manifold and connection logic

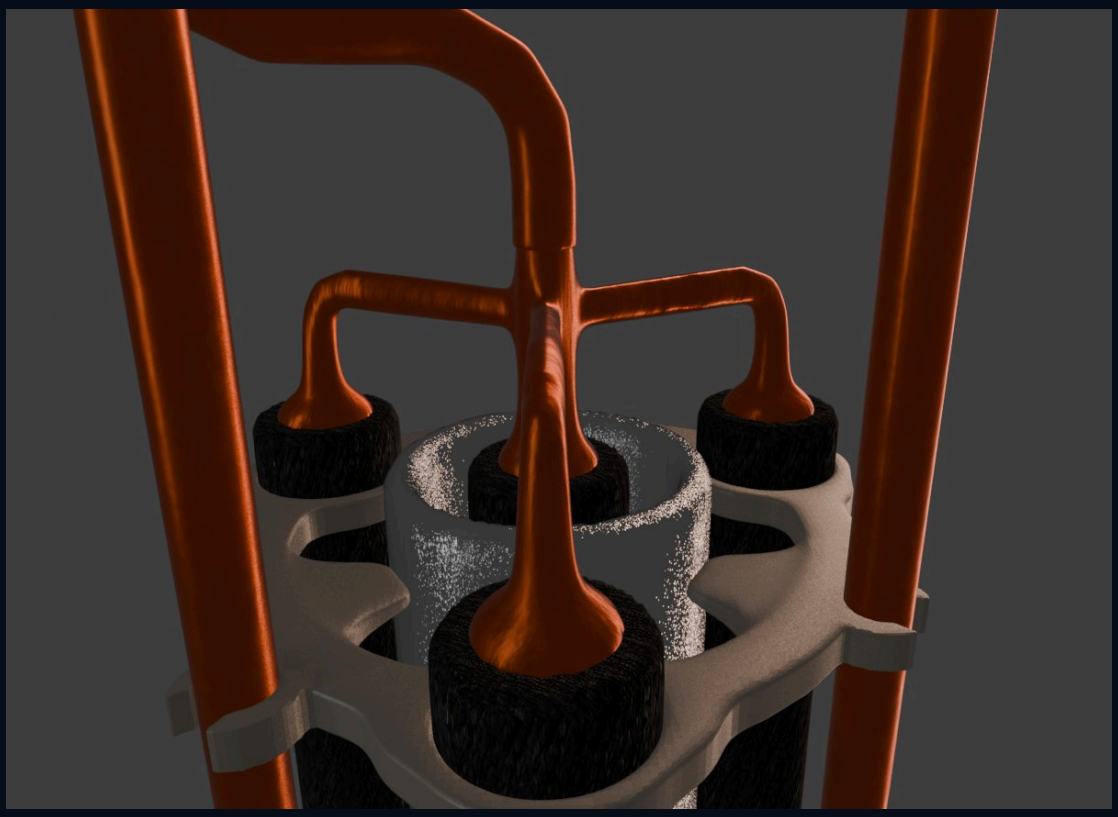


Fig. 4 — Full modular column stack. Multiple cells stacked vertically — this is the "fragment mesh" concept: each cell is

independent, replaceable, and connectable via the pipe-corner system.

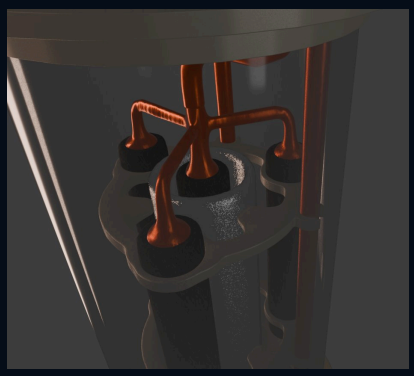


Fig. 5 — Electrode grid perspective

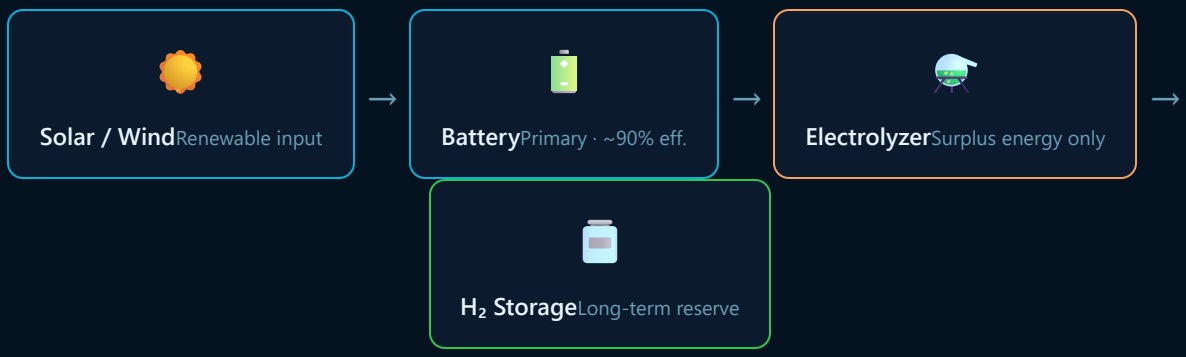
Fig. 6 — Under-slung perspective showing connection complexity

Fig. 7 — Top perspective render: anode/cathode alignment

3. CORRECT SYSTEM ARCHITECTURE

The architecture that works — the one that obeys physics — uses hydrogen as a **secondary storage layer**, not the primary energy path.

THE VALID ENERGY FLOW



Battery → Motor is the primary efficient path | Surplus → Hydrogen is the secondary buffer layer

Fig. 8 — Anode/cathode alignment. The anode surface fully encircles the cathode, maximising field uniformity and eliminating dead zones.

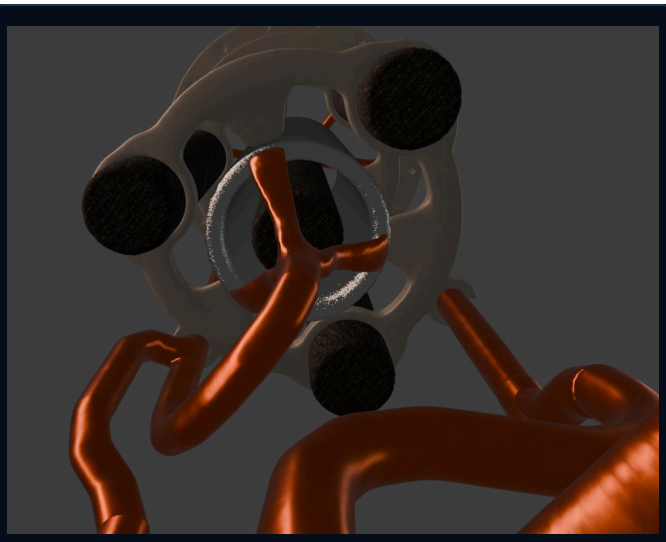


Fig. 9 — 120° tri-symmetric wireframe. Equal spacing prevents localised electrode degradation and distributes current density evenly.

⚠ CRITICAL SCALING RULE

Adding more electrolyzer modules does **not** increase total hydrogen output unless you also increase the electrical power supply proportionally. Output is always limited by available input power — not by how many cells are physically connected. This is an electrical scaling problem, not a plumbing problem.

4. TECHNICAL SPECIFICATIONS

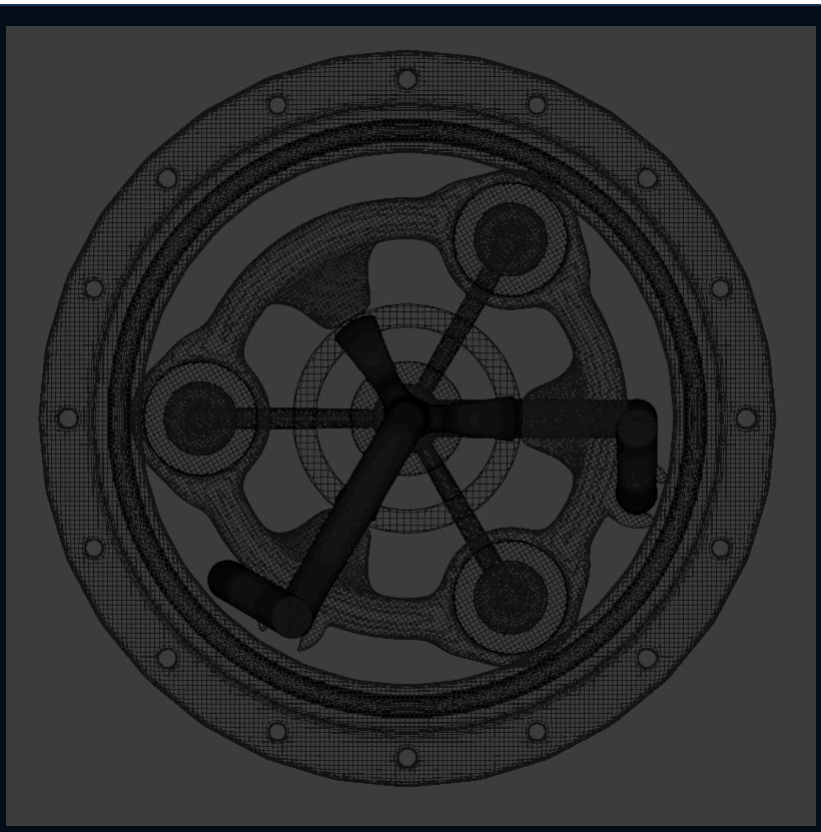


Fig. 10 — Internal electrode assembly. Carbon anode rods (dark) and nickel cathode pipe (metallic centre) inside the 60 mm glass cylinder filled with 20% KOH solution.

CYLINDER & ELECTROLYTE

PARAMETER	VALUE
Cylinder material	Borosilicate glass
Length	300 mm
Interior diameter	60 mm
Electrolyte	Potassium Hydroxide (KOH) — 20% by weight
Air gap	None — fully filled

ELECTRODES

PARAMETER	ANODE (CARBON)	CATHODE (NICKEL)
Material	Carbon — recycled battery rods	Nickel pipe

Quantity per cell	4 active (12 total)	**3 pipes**
Diameter	5 mm rod	**12 mm OD / 10 mm ID**
Length	55 mm	**55 mm**
Surface area	903.21 mm^2 per rod · 36.13 cm^2 total (4 rods)	**38.70 cm^2 per pipe**
Arrangement	120° circular spacing	**Central + peripheral**
Electrode gap	**2.5 mm between anode and cathode surfaces**	

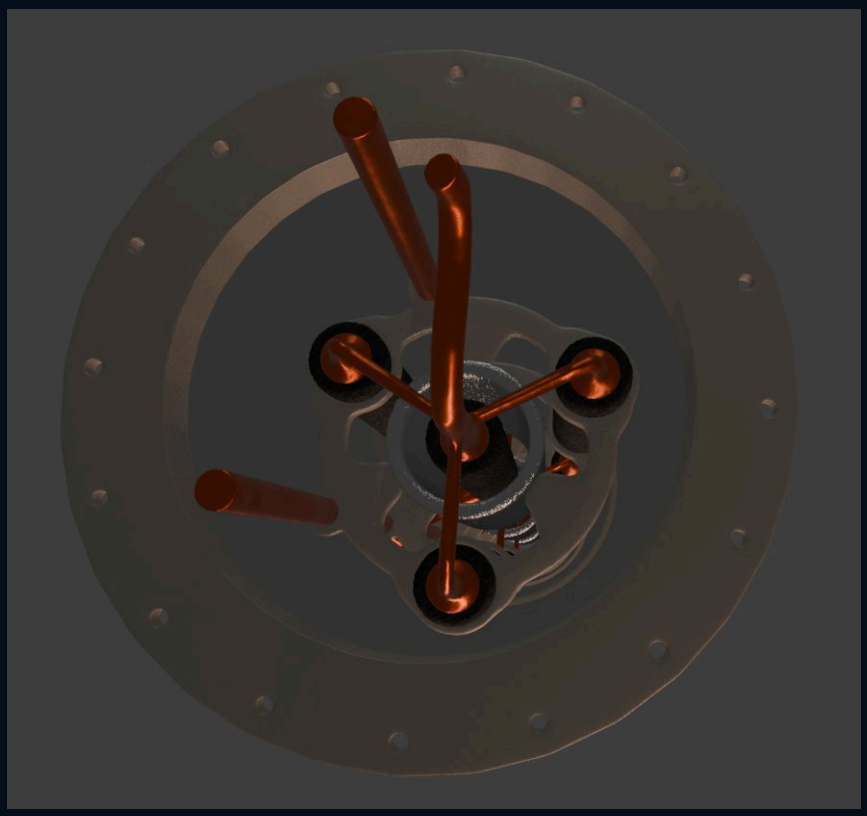


Fig. 11 — Bottom manifold and connection logic. The pipe-corner system allows cells to be linked horizontally or vertically. Gas is collected through single-direction valves — hydrogen and oxygen are never mixed.

GAS COLLECTION

PARAMETER	VALUE
Wiring	**Copper with chemical-resistant insulation**
Capture method	**Single-direction valve locking system**

Gas safety	No H_2 / O_2 mixing — separated collection zones
Output mode	Measured doses between valves
Scalability	Pipe-corner connectors allow horizontal and vertical cell expansion

5. HYDROGEN OUTPUT — FARADAY CALCULATION

```
// Total active anode surface area Surface = 36.13 cm² // Safe operating current density Current density = 150 mA/cm² // Maximum safe operating current I_max = 36.13 × 0.150 = 5.42 A // Hydrogen moles per second (Faraday's Law, n = I / 2F) n_H₂/s = 5.42 / (2 × 96,485) = 2.81 × 10⁻⁵ mol/s // Volume per second at STP (22.4 L/mol) V/s = 2.81 × 10⁻⁵ × 22.4 = 0.00063 L/s // Volume per hour — per single cell V/h = 0.00063 × 3600 = ≈ 2.27 L H₂ / hour // Energy content (1 L H₂ ≈ 3 Wh) Energy = 2.27 × 3 = ≈ 6.8 Wh per hour per cell
```

Scale context:

Single cell output (6.8 Wh/h) is laboratory / field-backup scale. For a ship's auxiliary load (~50 kW), approximately 26,000 cells and a matched power supply are required. Scaling is an electrical engineering challenge — not a physical impossibility.

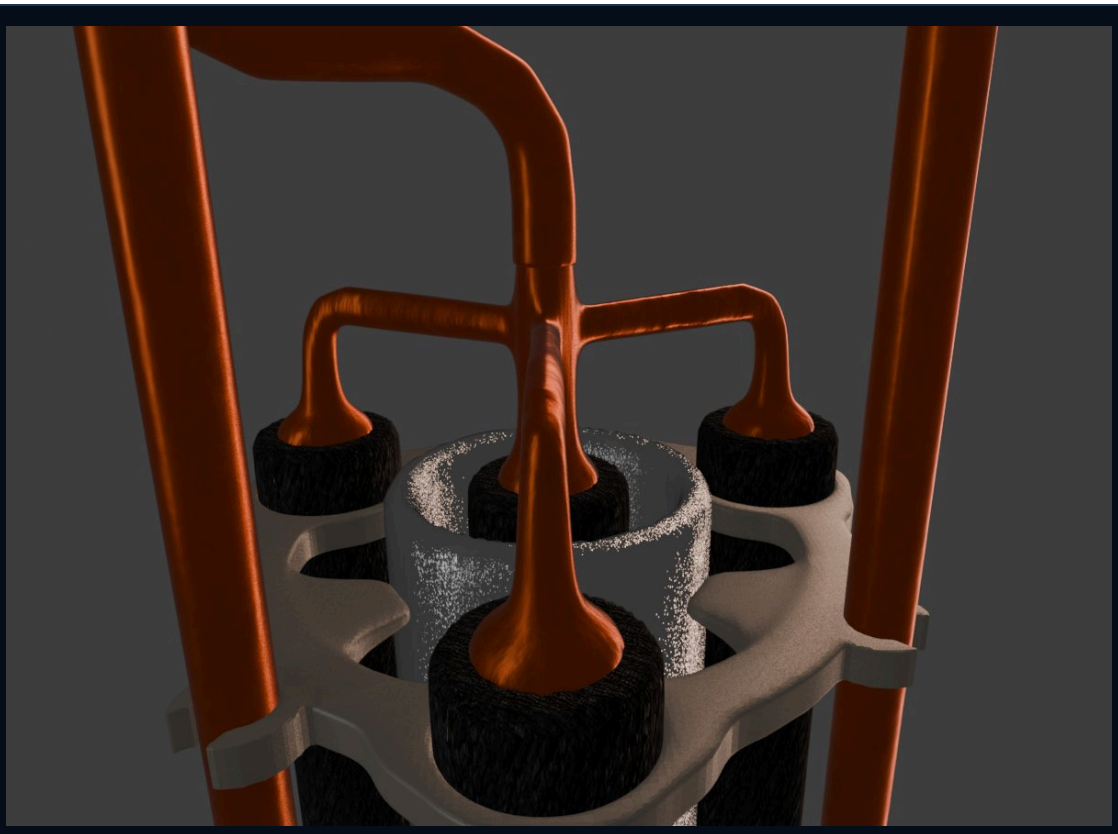


Fig. 12 — Full modular column stack. Each added cell multiplies output proportionally — but only if the electrical input is also scaled. The geometry stays identical at any size.

6. WHAT IT IS GOOD FOR

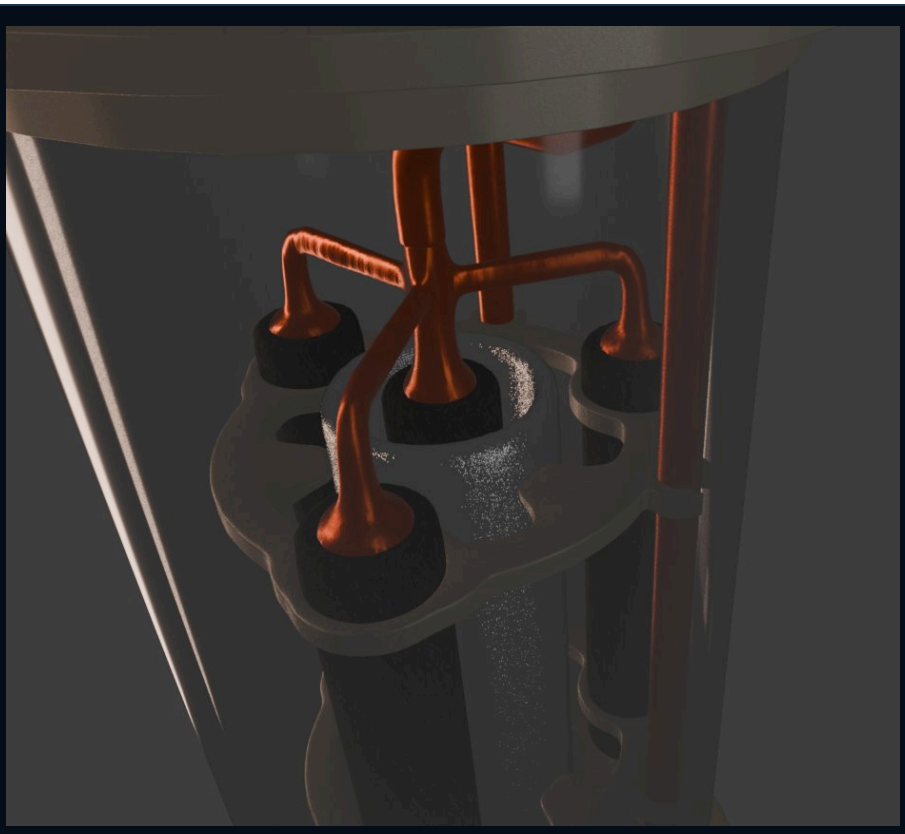


Fig. 13 — Electrode grid perspective. The 120° arrangement is not aesthetic — it is engineered for even current distribution, longer electrode life, and uniform gas production across the entire reaction zone.

🔋

Long-term energy storage

Hydrogen does not self-discharge. Store energy for weeks or months. Batteries cannot match this.

☀️

Solar / wind overflow buffer

When battery is full, convert surplus renewable energy into hydrogen instead of wasting it.

🏕️

Off-grid backup power

Remote cabins, field stations, small islands — store a hydrogen reserve when no grid is available.

🚨

Emergency reserve

When batteries are depleted and no charging source is available, hydrogen provides a fallback.

🛸

Lightweight energy (drones / field)

H_2 stores ~33 kWh/kg vs ~0.25 kWh/kg for lithium. Where weight is critical, hydrogen wins.

🔬

Research & education

Demonstrates Faraday's Law, electrochemistry, and electrode geometry effects in a fully buildable format.

🔥

🧩

Clean portable fuel

Burn hydrogen for cooking or heating in field situations. Only exhaust: water vapour.

Modular fault-tolerant design

One cell fails — others keep running. Individual cell replacement without stopping the system.

Small vessel auxiliary power

Yachts, fishing boats, tourist vessels — lighting, sensors, radio, refrigeration. ~100–200 cells.

Open-source green technology

Recycled carbon rods, affordable nickel, no rare earth metals, no patents. Free to build anywhere.

7. WHAT IT IS NOT GOOD FOR

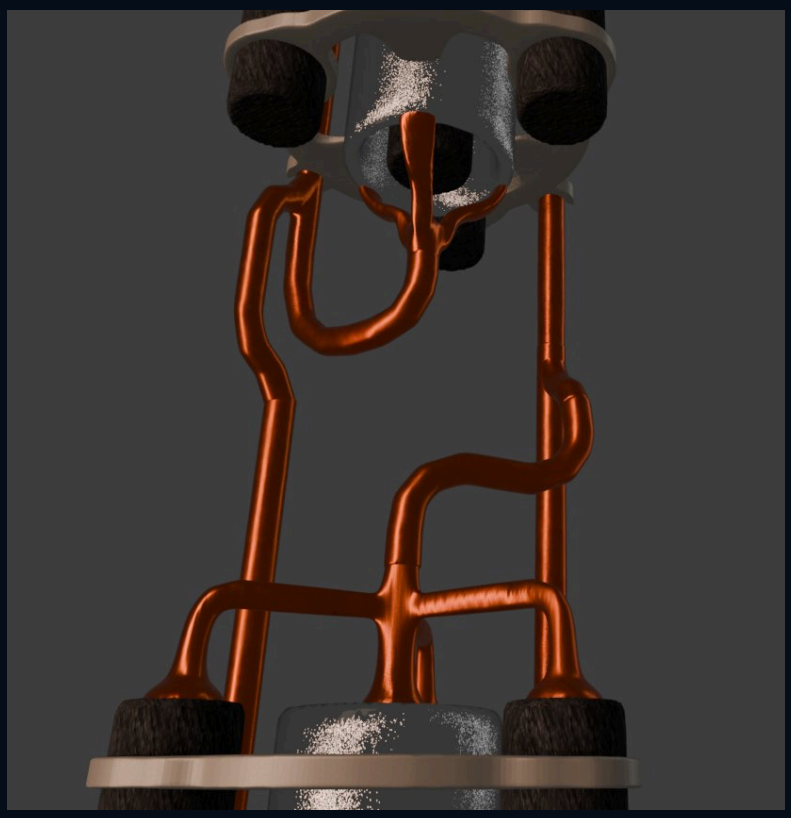


Fig. 14 — Under-slung perspective showing connection complexity. This is real engineering: valves, manifolds, electrical isolation, gas separation. Not a "plug and play" solution.

Energy amplification

You always get less energy out than you put in. No geometry or module count changes this. Conservation of energy is absolute.

Self-sustaining loops

No closed loop can power itself. You cannot take energy from motion to feed the same motion. Physics is non-negotiable.

🚗

Vehicle propulsion efficiency

Batteries are 2–4× more efficient for daily driving. Adding hydrogen steps reduces usable range, not increases it.

💸

Cost savings without solar

If input electricity is paid for (grid), hydrogen costs more per km than using a battery directly.

🚢

Large ship main propulsion

Main engines need megawatts. Solar on a ship delivers kilowatts. The gap is orders of magnitude, not percentages.

💧

Freshwater source for crew

At this scale, fuel-cell water output is negligible — a few millilitres per hour per cell. Not a practical water source.

8. MARITIME APPLICATION — HONEST ASSESSMENT

Ships have genuine advantages: large surface area, access to desalinated seawater for electrolyte, and existing auxiliary power infrastructure. The modular fragment design is well-suited to maritime environments. But the scale gap must be understood clearly.

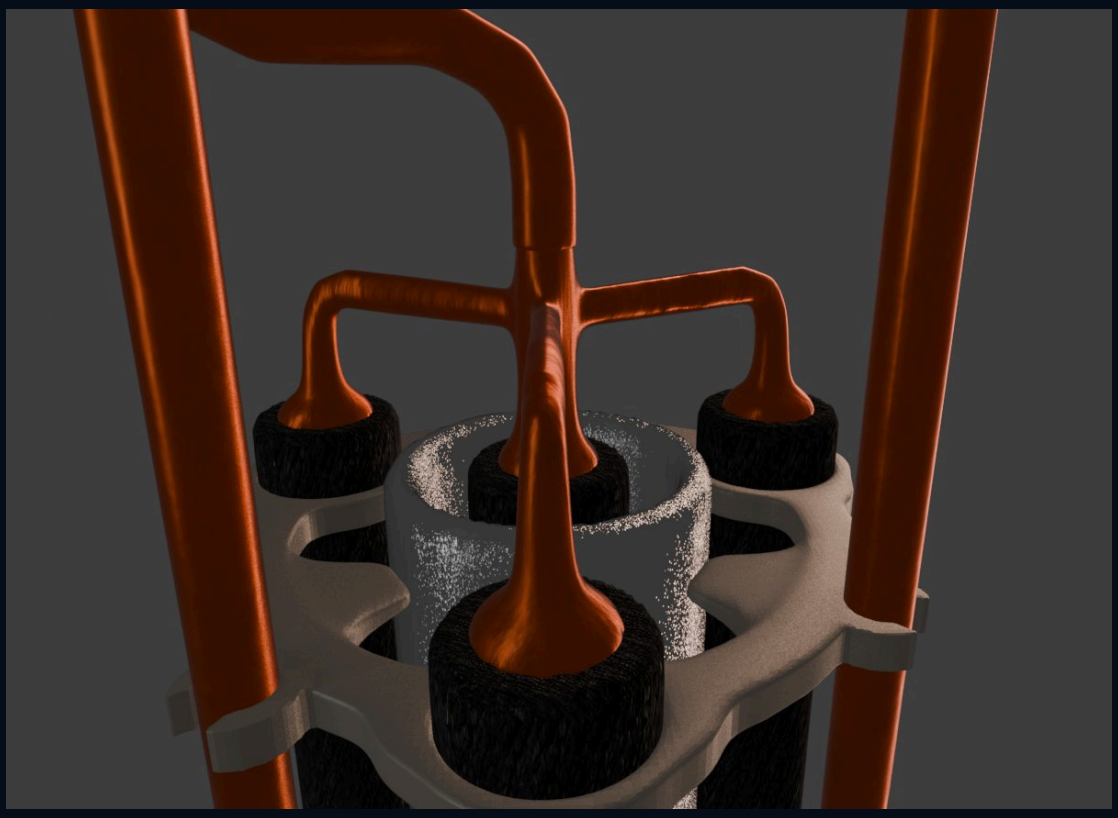


Fig. 15 — The modular column stack is the foundation of the maritime "fragment mesh" concept. Each column is an independent unit. On a vessel, hundreds of columns can be mounted in available spaces — hull compartments, deck edges, below-deck chambers — each contributing to a shared hydrogen reserve.

SHIPS VS CARS — WHAT CHANGES

FACTOR	CAR	SHIP (CARGO / FERRY)
Usable solar area	~5 m^2	**500–3,000 m^2 (realistic, not total deck)**
Solar energy per day (Mediterranean)	~10–25 Wh	**2,000–15,000 kWh**
Main engine power	50–200 kW	**5,000–50,000 kW**
Auxiliary engine power	N/A	**200–2,000 kW**
Water source	Carried	**Desalinated seawater on board**
Weight tolerance	Very limited	**High — ballast flexibility**

⚠ THE SCALING REALITY

A ship's auxiliary engine (~50 kW) requires ~60,000 L H_2/hour to replace. Your single cell produces 2.27 L/h. That means ~26,400 cells running continuously — each needing its own electrical supply, gas manifold, and maintenance access. This is technically achievable as an industrial system, but it is not a simple mesh extension. It is a full engineering project that requires matched solar or grid input power.

MARITIME USE CASE TABLE

APPLICATION	FEASIBILITY	CONDITION
Small boat auxiliary (lights, radio, fridge)	✓ Realistic	**100–200 cells with matched solar input**
Port / harbour zero-emission manoeuvre	✓ Realistic	**Pre-stored H_2 from shore or solar charging**
Greek island backup power station	✓ Realistic	**Modular ground installation, surplus solar**
Ferry / yacht auxiliary support	⚠ Marginal	**Viable with large solar array and proper storage**
Cargo ship auxiliary power	⚠ Marginal	**Technically possible, but costly vs. LNG or battery alternatives**
Large ship main propulsion (full replacement)	✗ Not realistic	**Energy gap is orders of magnitude — not a geometry problem**

WHERE IT GENUINELY HELPS THE GREEK FLEET

→ **Auxiliary loads** on ferries and fishing vessels using onboard solar H_2
→ **Port emission reduction** — switch to stored H_2 near islands (Santorini, Mykonos, Rhodes)
→ **Island microgrids** — same modular cells used on ships can power island communities from solar surplus
→ **Training platform** — Greek engineers learn hydrogen systems on low-cost, open-source hardware
→ **Green shipping certification** — even partial H_2 integration improves EU compliance and commercial ratings

Version 1.0 · Based on MCF 300 mm cylinder, KOH electrolyte, carbon/nickel electrodes, 2.27 L/h per cell

SECTION A — PURPOSE

This document defines what the MCF Hydrogen Mesh system can do, what it cannot do, and how it should be correctly understood. Its goal is to prevent confusion between energy storage, energy generation, and energy conversion systems.

SECTION B — ENGINEERING BOUNDARY STATEMENT

NON-NEGOTIABLE PHYSICS

The Hydrogen Mesh does not create energy. It only stores and transfers energy between forms. All performance depends on available electrical power, system efficiency, and thermodynamic losses. No configuration of geometry, pipes, or modular expansion changes these limits.

SECTION C — CORRECT SYSTEM ARCHITECTURE

Solar / Wind → Battery (primary, efficient) → Electrolyzer (surplus only) → H_2 Storage → Fuel Cell / Burner (backup use)

This is a two-layer storage system. Not a self-generating system.

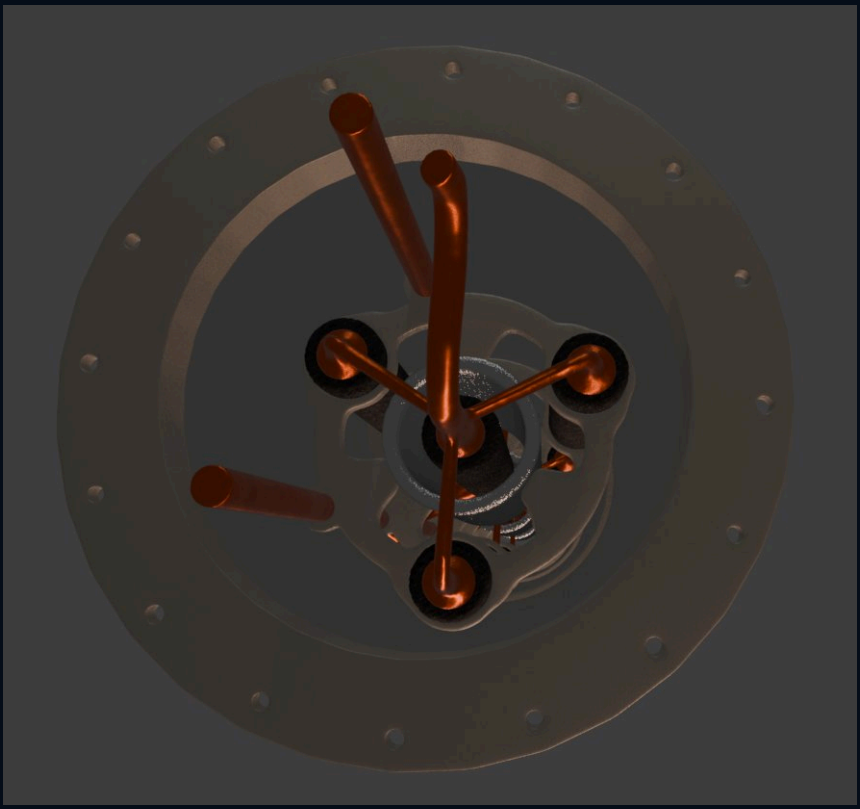


Fig. 16 — Manifold and valve connection system. Single-direction valves prevent backflow. Gas separation zones keep hydrogen and oxygen isolated. Copper wiring is insulated to prevent short-circuits in the KOH environment.

SECTION D — SAFETY NOTES

- KOH electrolyte is caustic — handle with gloves and eye protection at all times
- Hydrogen is flammable — all storage must use safety relief valves in ventilated spaces
- Gas separation must be maintained — H_2 and O_2 must never mix in enclosed spaces
- Electrical isolation is essential, especially in marine (saltwater) environments
- Glass cylinders are rated for atmospheric pressure only — not for compressed hydrogen storage

SECTION E — CLOSING STATEMENT

This system is meaningful when used within its physical limits. Its value lies in **resilience, flexibility, long-term storage, and distributed design** — not in energy amplification. Respecting these boundaries is what makes the design scientifically valid and practically useful. It is a stepping stone in green energy, not the final answer — and that is enough to be genuinely valuable.

10. LICENSE & ATTRIBUTION

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GitHub Repository

github.com/ChristosSotirelis/The-Morphean-Cosmology-Framework-MCF-

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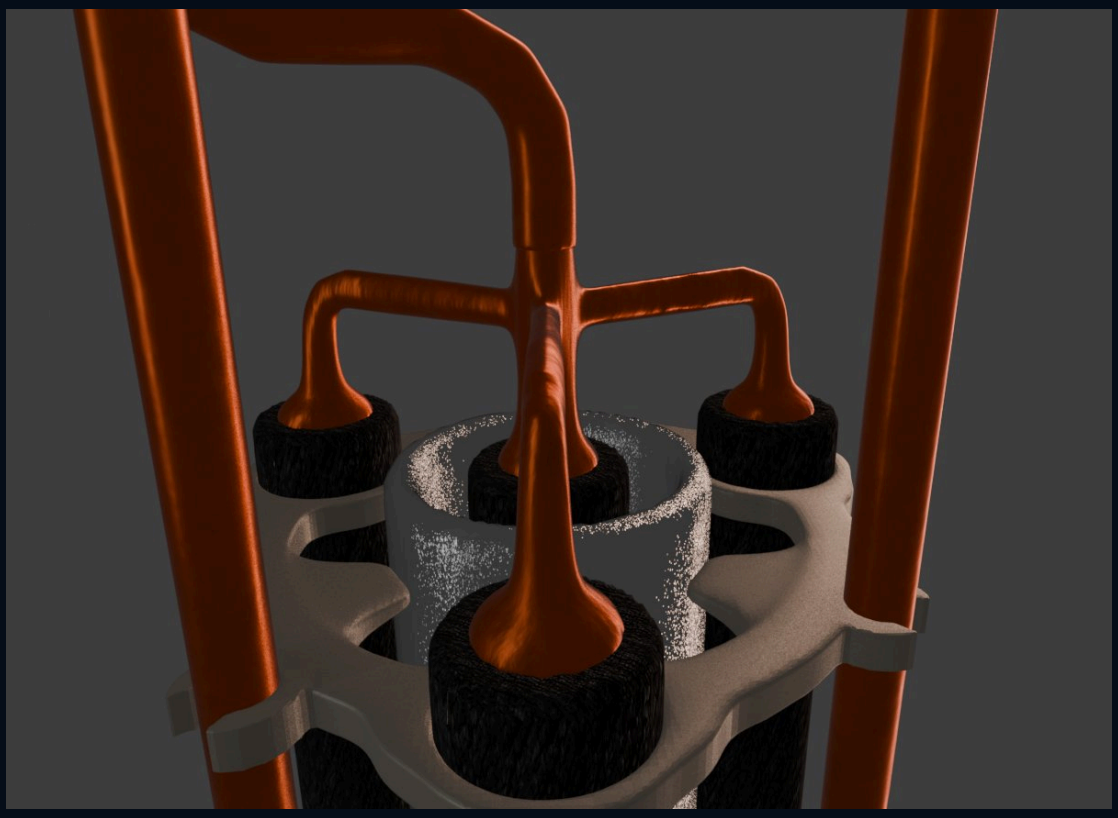

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